

GL-ProQ-en-017

Process Failure Modes Effects Analysis (PFMEA) Procedure

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Life Is On



Purpose

- The purpose of this procedure is to establish a clear and methodical approach for executing Process Failure Modes & Effects Analysis (PFMEAs) and effectively communicating process-related risks throughout Schneider Electric.
- PFMEA's evaluate potential failures in manufacturing, assembly, and logistical processes to ensure that the resulting products conform to the design intent.
- The PFMEA team analyzes processes by identifying potential failure modes stemming from process variation. This analysis is crucial for prioritizing actions to prevent the occurrence of potential defects which could reach customers. Additionally, it aims to enhance controls early in Industrialization phase to prevent unwanted defects caused by manufacturing or assembly, thus avoiding the resulting consequences for both internal and external customers.

Scope

- This PFMEA procedure **shall** be applied to all Schneider Electric **Offers**, ensuring comprehensive assessment and mitigation of potential process-related failure modes across the entire product and solution portfolio. This includes the thorough process risk evaluation of pertinent process steps, and process elements and its relation to product characteristic(s) to uphold the highest standards of safety, quality, cost, and delivery to customers.

Target Audience

- All relevant cross-functions involved in PFMEAs at Schneider Electric

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References

- GL-ProQ-016 DfSR DFMEA Procedure
- GL-ProQ-022 CTQ_CTP Management Procedure
- GL-ProQ-033 Process Capability Study and Analyze Results Procedure
- GL-ProQ-034 Control Plan Procedure
- GL-DirQ-043 Quality Fundamentals for GSC
- GL-ForQ-033 PFMEA Template
- GL-GuiQ-027 PFMEA O/D FAQ (Frequently Asked Questions)
- GL-Gui-028 PFMEA Scoring rules applied to failure modes

Process Inputs

- Offer Functional CTQ Characteristic List (ref. GL-ProQ-016 DfSR DFMEA Procedure)
- System, Subsystem and/or Component DFMEA (ref. GL-ProQ-016 DfSR DFMEA Procedure)
- Product Characteristic(s)/Characteristics CTQs (ref. GL-ProQ-016 DfSR DFMEA Procedure)
- Process Flow Diagram
- Process Assumptions
- Plant Layout
- Field performance and quality data
- Lessons learned from prior products
- Project goals (ppm, reliability, etc.)

Process Overview

At Schneider Electric, PFMEAs adhere to the 7-step process flow detailed in Figure 1 to achieve its process outputs. Teams are encouraged to embrace an agile approach and manage process risks though PFMEAs, which align with the Offer objectives and timelines established by the team. However, PFMEAs **shall** fulfill the **intended outputs** of each of the 7 steps. To ensure this, it is suggested that the process steps be carried out sequentially, as the activities in each step build upon those of the previous step.

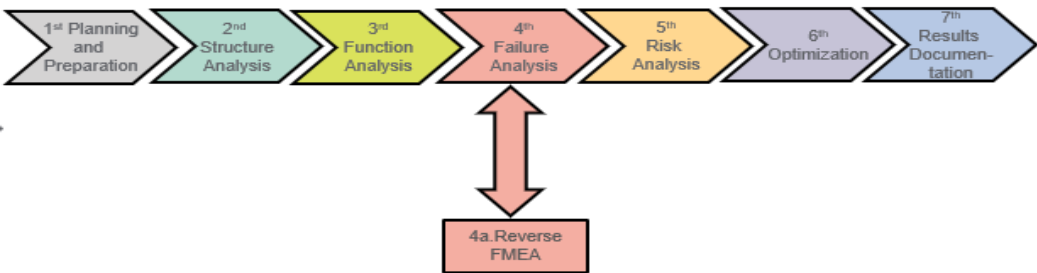


Figure 1: Schneider Electric PFMEA process flow

Step 1: Planning & Preparation

Follow the steps below to complete **Step 1: Planning & Preparation**.

1) Define Scope

PFMEAs can include hundreds of operations and be resource intensive. The IQL (Industrial Quality Leader) or someone with an equivalent competency profile **shall** be responsible for ensuring the intended outputs of all the PFMEA process steps are completed. The IQL **shall** have established an Expert level competency rating, recognized through the Schneider Electric **CoMET** (Competency Management) system or through approval from authorized LoB (Line of Business) PFMEA SME. There are 3 (three) cases at Schneider Electric the organization **shall** consider when defining the scope of a PFMEA.

Case	Description
Case 1 – New Offers	A new process, test, plant layout and/or processing technology; PFMEA scope is the complete process flow linked with the system, subsystem, and/or component from the associated DFMEA (mandatory input). At a minimum, all process steps linked with Functional and/or Characteristic CTQs flown down from engineering and/or DFMEA Failure Mode Effects with a Severity ≥ 7 shall undergo a PFMEA analysis.
Case 2 – I2P	Modifications to an existing process; scope is the process modifications, interactions with unchanged portions of the process, and performance history (Safety Incidents, 8Ds, customer complaints, warranty, recalls, test failures, scrap, rework, etc).
Case 3 – Plant Led Industrialization	Use of an existing process for a different application or environment (facility); scope is the location and environment factors.

2) Intended Outputs

The following, at a minimum, **shall** be defined prior to PFMEA kickoff, the Product Owner listed as a sponsor (Case 1) and Plant Manager listed as a sponsor (Case 2 & Case 3).

Step 1: Planning & Preparation (cont.)

The IQL, or function with equivalent competency is responsible for defining the following:

Key considerations	Description
InTent	Scope of the PFMEA, in accordance with the 3 cases
Team	The PFMEA team must be cross-functional, consisting of individuals with relevant expertise in manufacturing, assembly, or service operations. All members should be trained in the Schneider Electric PFMEA methodology. The PFMEA Leader (The IQL or equivalent Plant Quality Engineer at GSC/ETO/DC sites) is responsible for forming the team based on available resources, project scope, timeline, defined EPICs, and budget. Team composition should include: Core Team • SMEs directly involved in the process being analyzed (e.g., process engineers, quality engineers, process owners). • Members with deep knowledge of the design, product, process flow, and associated risks. • Actively participate in all PFMEA steps and decision-making. • Remain engaged throughout the entire PFMEA lifecycle to ensure continuity and accountability. • Aligned to the RACI chart defined in the Roles and Responsibilities section of this procedure. Extended Team • Stakeholders who provide input or support at specific stages (e.g., logistics, maintenance, supplier quality, operators, service teams). • Engaged as needed based on process complexity, product scope, or specific risk areas.
Timing	PFMEAs are executed through a series of meetings. The objectives of the meetings are based on the timeline for the PFMEA completion, which shall be defined to achieve the scope of the analysis. Note: For Case 1, the PFMEA timing should be in-line with both the DFMEA system/subsystem/ component level timing and integrated with key Offer objectives and timelines as part of OLM. A recommended tool to define the timing is a Gantt chart.
Task	It is the responsibility of the IQL that the cross-functional team comes prepared for every PFMEA meeting, per the project timing. There shall be an evaluation of existing inputs available for the PFMEA as well at defined intended outputs. Focus on inputs which include but is not limited to, pre-existing PFDs (Process

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	Flow Diagrams), Drawings, Model files, Work Instructions (similar parts & processes), Quality performance data (similar parts & processes), BOM (Bill of Materials), DFMEA reports, CTQ lists, Control Plans (similar parts and processes). Note: Generic FMEAs shall be used as inputs, based on (1) availability and (2) scope of the PFMEA process under evaluation (i.e. Welding process). Generic FMEAs contains both historical and potential failure modes, causes, controls, etc that are incredibly valuable during Planning & Preparation. However, Generic FMEAs are not related to the specific product in development and shall only be used as an input which is updated and linked to specific product characteristic(s) for the specific offer/solution.
Tool	The infrastructure to complete the PFMEA shall be verified and defined by the project team. The IQL has the autonomy and empowered to define the PFMEA infrastructure that will be followed as part of this step. Considerations should include the current competencies of the team, which can be determined at the discretion of the IQL. At SE, the PFMEA team shall utilize the standard Excel based template that can fulfill the intent of the PFMEA 7-step process outputs (ref. GL-ForQ-033 PFMEA Template). <i>Note 1: ePFMEA aligns to GL-ForQ-033 PFMEA Template, so usage of ePFMEA aligns to this Tool requirement.</i>

Step 2: Structure Analysis

Ensure that Intended Outputs are met to complete **Step 2: Structure Analysis**.

Intended Outputs	Description
Process Step	- The PFMEA team shall construct a Process Flow Diagram (PFD) on the process under evaluation, based on (1) Defined Scope and (2) Process Item. - The PFD is a visual representation of the process steps as it presently exists when “walking the process.” In other words, the PFD highlights the specific manufacturing station where the process functions are applied to the system, subsystem and/or component to achieve the desired product characteristic(s). - The team can elect to use an excel based Schneider Electric PFD template to assist in documentation of the process steps. Example: “Machining Station” Note: All Product Characteristics/CTQs shall be allocated to the Process Step Level, not for each Function (sub-step/operation)

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Process Work Element	- The Process Work Element is the lowest level of the Structure Tree. - Each work element is the name of a main category of potential causes of failure modes that could impact the process step. PFMEA teams shall consider 6Ms for each work element of each process step (<u>M</u>achine, <u>Hu</u>Man, <u>M</u>aterial, <u>Environ</u>Ment, <u>M</u>ethod, <u>M</u>easurement). Note 1: The 6Ms are intended to be “thought starters” for the PFMEA team to stimulate deeper insights and thoroughly analyze the process steps. It is therefore not required to categorize each “M,” but rather mandatory for the team to consider each “M” and then collectively decide what is appropriate to document in the PFMEA. Note 2: Each work element delves into the specific tasks or actions within each process step and the functions they serve. Think of it as all the required “ingredients” in the process step to achieve the intended product and process characteristic(s). Example: CNC Machine (Machine), Machining Operator (HuMan), Drill Bit (Material), Coolant (Material)
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Step 3: Function Analysis

Ensure that Intended Outputs are met to complete **Step 3: Function Analysis**.

Intended Outputs	Description
Function (Sub-step/Operation)	“Verb-noun” explanation of basic operation Example: Drill hole #1
Product Characteristics (Requirements)	- A Product Characteristic refers to any distinctive feature or attribute of a product that defines its quality, performance, and/or functionality requirement(s). These characteristics can include, but are not limited to, physical dimensions, material properties, functional specifications, and performance parameters that are essential for the product to meet its intended purpose and fulfill customer requirements. - Product characteristics shall be linked directly to the associated process step. Examples include, but not limited to, length, depth, surface finish, pre-load, etc. - Product Characteristics listed shall be quantitative (when possible). Sources for Product Characteristic(s) include, but are not limited to, DFMEA functions, product drawings, reliability, maintainability and availability requirements, test requirements, software requirements, and/or engineering specifications.

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	• Example: Hole #1 Diameter of 50 mm +/- 0.2 mm (product characteristic) • Note 1: There can be multiple product characteristics associated with each process step and should be listed separately. • Note 2: The CTQ list flown down from Engineering, shall be distinguished in the PFMEA scope defined in Step 1: Planning & Preparation (Schneider Electric excel based template).
Process Characteristics (Requirements)	• Process Characteristics refer to the specific attributes and parameters of a manufacturing process step that directly influence the ability of the process step to achieve the required Product Characteristics. • Process Characteristics shall be quantitative (when possible) and directly linked to Work Elements defined in Step 2: Structure Analysis. Sources include, but are not limited to, manufacturing drawings or specifications (including operator work instructions, machine specs, set-up instructions, error proofing verification procedures, etc.). • Examples: Drilling speed (Method), Coolant Pressure (Method), Tool Maintenance Schedule (Machine), Temperature (EnvironMent). • Note 1: There can be multiple process characteristics associated with each product characteristic and should be listed separately.

Step 4: Failure Analysis

Ensure that Intended Outputs are met to complete **Step 4: Failure Analysis**.

Note: Failure Analysis relies on creating a Failure Chain. This is a unique combination of failure cause, failure mode and failure effect which is generated because of the Failure Analysis step.

Intended Outputs	Description
Potential Failure Mode	• A Potential Failure Mode is defined as the way the process step could potentially fail to meet the defined product characteristic(s) (defined in Step 3: Functional Analysis) • There are several “thought starters” when brainstorming potential failure modes, including, but not limited to: - Does not meet function - Over-achieving function (i.e. operation output above acceptable threshold) - Under-achieving function (i.e. operations output under acceptable threshold) - Intermittent function (i.e. operation output randomly in and out of the specification) - Unintended function (i.e. operator safety)

	• Note 1: The DFMEA has critical information that the PFMEA team <u>shall</u> consider when completing the analysis of potential failure modes. • Note 2: There can be multiple potential failure modes associated with each functional requirement and should be listed separately.
Potential Effect of a Failure Mode	• A Potential Effect of a Failure Mode is the consequence or result of each potential failure mode on the end user and <u>shall</u> be listed in the PFMEA. • External Effects for PFMEA <u>shall</u> be linked to the standard End-Effects list from ref. GL-ProQ-016 DfSR DFMEA Procedure • <u>External Effects (Customer/End-User)</u> - Product Operation - Final Customer - Compliance with safety regulations • Note 1: There can be multiple potential end effects associated with each potential failure mode and should be listed out separately. • Note 2: At a minimum, all external effects of failure modes with Severity ratings ≥ 7 from the associated system, subsystem and/or component DFMEA <u>shall</u> be directly transferred into the associated system, subsystem and/or component PFMEA. • Note 3: Internal Effects (i.e. Operator Safety, Downstream users and Machine/Equipment), are not covered in PFMEA, but <u>shall be</u> evaluated using the Pilot Run/PPAP IG process, as the focus on the PFMEA is only on External Effects to align with OLM. For example, Operator Safety risks should be addressed through safety audits from Pilot Run/PPAP process. • Note 2: The only exception to the External Effects of the Failure Mode being derived from GL-ProQ-016 DfSR DFMEA Procedure is PFMEAs from Distribution Center/Factory Warehouse, which can align to the External Effects noted in Appendix A
Potential Cause of a Failure Mode	• A potential cause of a failure mode is the reason why the potential failure mode could occur. The consequence of a cause is the failure mode. The cause is directly correlated to the Process Work Element defined in Step 2: Structure Analysis, which is the lowest level of the process hierarchy. • At Schneider Electric, potential causes of failure modes are defined by evaluating the 6M causes: - <u>Machine</u> (machine capability, initial setup adjustment, machine wear over time, inadequate gating/venting, inadequate or no lubrication, tool wear over time, tool breakage, tool-to-tool differences, fixture tolerance, fixture adjustment, fixture wear over time, chip on locator, worn locator, weld current too high or low, weld pressure, heat treat temperature too high or low, conveyor speed too fast or too slow, equipment maintenance including repair, replacement, reassembly, and

	adjustment, inspection gauging failures including inaccuracies, and ineffectiveness, etc) - Methods (sequence, procedures, layout, off-line rework/repair, off-line inspection, material flow, process control programming, scrap containment, etc.) - Material (part missing, part mislocated, incoming raw material, purchased parts, previous operations) - HuMan (manual over torque, manual under torque, operator skill, ergonomic factors, time, missing or inadequate visual aids, lack of concentration, etc.) - Measurement (gauge wear out, gauge out of calibration, etc) - EnvironMent (e.g. plant temperature, lighting, humidity, dust, noise, etc.) • Note 1: <i>There can be multiple potential causes of failure modes associated with each failure mode and should be listed out separately.</i> • Note 2: <i>“Human Factors” and associated variations (i.e. Operator Error) shall not be listed as a potential cause of a failure mode. Instead, the PFMEA team should consider Human Factor cause types such as Workstation Layout, Ergonomics, Documentation/Training and/or Cognitive/Attention/Concentration as Potential Causes of Failure Modes.</i> • Note 3: <i>It is recommended that P-Diagrams be utilized to help brainstorm causes of failure modes.</i> • Note 4: <i>The PFMEA team, shall consider the 7 common causes of failure modes for assembly-related process steps (missing parts, wrong parts, wrong assembly, wrong direction, non-function, broken component, under-assembly).</i> • Note 5: <i>The PFMEA team shall consider 3 common causes of failure modes for mislabeling (zoning, identification, and validation).</i>
Step 4a: Reverse FMEA	• Reverse FMEA is a mandatory validation method used during Step 4 of the PFMEA process. • It ensures the PFMEA reflects actual shop floor conditions and identifies any missed or emerging risks. • Reverse FMEA shall be conducted only after completing Steps 1–3 to ensure the analysis is grounded in a structured foundation and not done in isolation. • To complete a Reverse FMEA, the cross-functional PFMEA team shall go to the production floor in-person to observe the process and engage with operators real-time. • Teams shall verify in-person that documented failure modes, causes, and effects are still valid and relevant. • Additional risks may be identified through direct observation and operator feedback and shall be incorporated into the PFMEA analysis.

	• Completion of Reverse FMEA at this step ensures occurrence, and detection ratings are validated using real-world evidence. • Note 1: <i>If the process involves a new production line, teams should reference a similar or existing line to support the in-person Reverse FMEA evaluation.</i> • Note 2: <i>Teams can and are encouraged to begin Reverse FMEA observations as early as Step 1 to build process familiarity, but at a minimum it must officially start no later than Step 4. This ensures the activity is based on a structured and informed analysis—not conducted in isolation.</i>
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Step 5: Risk Analysis

Ensure that Intended Outputs are met to complete **Step 5: Risk Analysis**.

Intended Outputs	Description
Current Prevention Controls	• Prevention controls describe how a cause of a failure mode is prevented from occurring during the process. • Prevention controls shall be considered when developing a PFMEA and should be evaluated based on what's currently in place. • Prevention controls are based on the application of effective standards, specifications, process rules, process guides, lessons learned, etc. to either completely avoid the potential cause of the failure mode or reduce their rate of occurrence. • Example: <i>Poka-Yokes, Equipment maintenance (e.g. skilled trades performed), Operator maintenance, Work instructions/Visual aids, Machine controls (i.e. machine monitoring of fluid levels).</i> • Note 1: <i>There can be multiple prevention controls associated with each cause of a failure mode and should be listed out separately.</i>
Current Detection Controls	• Detection Controls shall be considered when developing a PFMEA and must be evaluated based on what's currently in place. • Detection controls describe how the potential failure mode and/or potential cause of failure mode, is detected based on automated and/or manual methods that are currently in place. • Detection controls assume the failure occurred and describes how the cause and/or failure mode can be detected prior to shipment to the customer. • Examples: Check-do-check, Setup verification (3-piece inspection), Variable measurement, Attribute measurement, Functional check, Visual inspection, Over-torque alarm, Outgoing material audit).

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	• Note 1: <i>There can be multiple current detection controls associated with each cause of the failure mode and/or failure mode.</i>
Assess Severity of Effects of Failure Modes	• Severity is a rating number associated with the most serious potential effect of a failure mode for the process step under evaluation. • PFMEA Severity values for <i>external</i> effects shall derive from the associated system, subsystem and/or component level DFMEA. Additionally, PFMEA Severities shall be evaluated independently in the PFMEA for <i>internal</i> effects • All PFMEA Severity values shall be evaluated using the Schneider Electric Severity Table (Appendix A) as it is aligned directly with ref. GL-ProQ-016 DfSR DFMEA Procedure • Severity values ≥ 9 shall require immediate action priority from the PFMEA team, prior to release. • Note 1: <i>There can be multiple Severity values associated with each potential effect of a failure mode.</i> • Note 2: <i>PFMEAs from Distribution Center/Factory Warehouse have a dedicated Severity column in Appendix A</i> • Note 3: <i>For specific end effect and severity values of different LoBs, please ref. GL-ProQ-016 DfSR DFMEA Procedure</i>
Assess Likelihood of Occurrence (Cause of Failure Modes)	• Occurrence is a rating number associated with each potential cause of a failure mode being evaluated. The occurrence rating considers the likelihood of occurrence during production given the current Prevention Controls only. • Occurrence values shall be evaluated using the Schneider Electric Occurrence Table (ref. Appendix B). • Note 1: <i>Assessing the Occurrence value requires a thorough analysis of existing data, including scrap, rework, and failure rates. This evaluation is crucial for accurately determining the Occurrence rating and identifying areas for preventive action.</i>
Assess Detection	• Detection is a rating to evaluate the ability of the current detection control to detect the failure mode and/or cause of the failure mode. • Detection values shall be evaluated using the Schneider Electric Detection Table (ref. Appendix C).
Assess Risk Priority Number (RPN)	• The RPN (Risk Priority Number) is the product of Severity * Occurrence * Detection. • RPNs are a key indicator used in PFMEA to quantify and prioritize process risks, guiding PFMEA teams in making informed decisions about where to focus mitigation efforts to optimize risk for robust offer safety, quality and operational performance. • At Schneider Electric, we use thresholds through Green, Red and Black classifications. These are used to prioritize actions. Actions with Severity ≥ 9 and Black RPNs ≥ 144 shall be completed prior to release (ref. Figure 3, Monitoring & Control)

Step 6: Risk Optimization

Ensure that Intended Outputs are met to complete **Step 6: Risk Optimization**.

Intended Outputs	Description
Develop Recommended Actions	- Development of recommended actions are based on prioritization from the established thresholds (ref. Figure 3, Monitoring & Control), and represents a commitment to take specific, measurable and achievable actions to optimize risk. - Prevention-type recommended actions shall reduce the likelihood of occurrence of a cause and its resulting failure mode and/or effect. Examples include, but not limited to, Poka-Yokes (error proofing), reducing variation (increasing Cpk/Ppk) and reviewing product design for process optimization. - Detection-type recommended actions shall increase the ability to detect the cause and/or failure mode at operation. Examples include Jidoka (self-correcting process step) and/or increased sensitivity of testing to exhibit the failure mode. Note 1: Temporary Quality Wall (TQW) shall be implemented as a containment for all Sev ≥ 9 and Black RPNs, that are awaiting actions to be completed. Note 2: Severities for product (external) effects, can only be lowered by the system, subsystem and/or component DFMEA team through requirements/function change.
Responsibilities of Actions	Each action shall have a responsible individual and targeted completion date.
Assessment of Action Effectiveness	- PFMEAs shall be treated as a living document, updated regularly as actions are completed by responsible individuals. - Whenever an action has been completed, Severity (as applicable), Occurrence and Detection values are reassessed to ensure sufficient Risk Optimization.

Step 7: Results Documentation

Ensure that Intended Outputs are met to complete **Step 7: Results Documentation**.

1) Intended Outputs

- The IQL **shall** summarize and communicate the results of the PFMEA to all key stakeholders identified in the RACI (see [Roles & Responsibilities](#) section).
- The PFMEA team **shall** summarize high risk failure modes and communicate to key management representatives.
- OSRC (Offer Safety Review Committee) **shall** review all Severity ≥ 9 and is empowered to mandate actions to the PFMEA team should they desire for all other Severity levels.
- The PFMEA team **shall** commit to using the PFMEA for continual improvement and to communicate effectiveness of lessons learned.
 - The PFMEA team **shall** support the Problem Leader of an 8D to update the relevant PFMEA as an output of Step 7 (Prevent Recurrence) of the 8D process.
 - The PFMEA team **shall** support the update of PFMEA as an output of a Reverse FMEA activity.
- Examples, include but are not limited to, revising PFMEA during mass production (Case 1), updating failure modes (Case 2), and updating during production transfers (Case 3).

The PFMEA team **shall** commit to linking the PFMEA with the DFMEA, Control Plan and Standardized Work through updated CTQs and/or CTPs.

Process Outputs

- Updated CTPs (ref. **GL-ProQ-022 CTQ_CTP Management Procedure**)
- Risk Mitigation Actions
- Process Capability Studies (ref. **GL-ProQ-033 Process Capability Study and Analyze Results Procedure**)
- Key elements of the Control Plans (ref. **GL-ProQ-034 Control Plan Procedure**):
 - Process-related Prevention and/or Detection Controls
 - Standardized Work Instructions
 - Sampling Plan
 - Inspection Methods
 - Inspection Tooling
 - MSA (Measurement System Analysis) Plans
 - Reaction Plans
 - Inspection Plans
 - Training Plans

Monitoring & Control

Severity	RPN Color by Severity			
		Black	Red	Green
	9&10	>=144	140 - 126	< = 100
	7		>= 196	< = 140
	4		>= 280	<= 196
	1			<= 100

Figure 2: RPN Matrix

	Guidance		Governance	
	Intention of Color	General guidance	OLM	Legacy (including industrial transfer)
B	S&R Risk	Mandatory Action	No Pilot Run and no Release if Black RPN are present. Validated by PPAP-IG & OSRC review.	Mandatory to include in risk reduction roadmap.
R	Quality risk is main priority. S&R risks exist, Delivery and Cost will benefit also	Highly Recommended Action	For S9/10 OSRC decision required. For all: Decision required according to OLM Chart of Authority.	Prioritized to include in risk reduction roadmap.
G	Low Risk / Opportunity	Team Decision		

Figure 3: Governance Guidance

- In additions to the above to manage RPNs and provide governance, the Global CS&Q function **shall** conduct annual PFMEA maturity assessments against each of the 7-process step intended outputs, on both new and legacy offers.

Roles & Responsibilities

As PFMEA development is a cross-functional activity, a **RACI** matrix as described in table below should be followed.

Function	RACI Roles
Industrialization Quality Leader (IQL) or similar	R
Product Owner	A
Customer Experience Developer/Offer Quality Leader/Project Quality Leader/DfSR Practitioner	C
Manufacturing Plant Quality	C
GSC Methods	C
Design Engineer	C
Scrum Master	I
League Leader	I

General

Competency Requirements

Role	Knowledge	Competency Level	Intervention
IQL	Advanced PFMEA Facilitation training & this procedure	Advanced	Advanced PFMEA Training provided by Regional and LOB Trainers and/or intervention by CoMET expert
PQL/OQL	This Procedure	Competent	Competent PFMEA training
Mfg. Plant Quality	This Procedure	Competent	Competent PFMEA training
GSC Methods/Indus Process Engineer	This Procedure	Competent	Competent PFMEA training
Design Engineer	This Procedure	Competent	Competent PFMEA training
Product Owner	This Procedure	Competent	Competent PFMEA training
League Leader	This Procedure	Basic	Basic PFMEA training
Scrum Master	This Procedure	Basic	Basic PFMEA training

Quality Records

Record	Record Location	Retention Period
Developed PFMEA	Applicable Offer Folder(s)	20 years

Terms & Acronyms

The table below includes definitions for terms and acronyms used in this document. Reference [ISEE](#) for a full list of terms and acronyms used in the SE IMS.

Term	Definition
PFMEA	Process Failure Modes & Effects Analysis
DFMEA	Design Failure Modes & Effects Analysis
CTQ/CTP S/R	Critical to Quality/Critical to Process Safety & Regulatory
CTQ/CTP Standard	Critical to Quality/Critical to Process Standard
RPN	Risk Priority Number
TQW	Temporary Quality Wall
LOB	Line of Business

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Appendix A: Severity Table

Severity			
S	Level	Failure Effect on (End) user / Customer for Hardware & Systems	Failure Effect on (End) user / Customer for Distribution center/Factory warehouse
1	Imperceptible	Undetectable by the Customer. Potential failure effect: no perceptible effect on product functionality <i>Customer is unlikely aware of it.</i>	Undetectable by the Customer. Potential failure effect: No perceptible effect on the delivery execution. <i>Customer is unlikely aware of it.</i>
4	Performance	Still working. Potential failure effect: some product degradation at the subsystem level, but performance of main systems is not affected. <i>Customer is made uncomfortable or annoyed.</i>	Still working. Potential failure effect: Secondary (nice to have) Delivery offer function degradation (Examples: Excess received by customer, Secondary/logistics packaging damage, Error on documentation/label). <i>Customer is made uncomfortable or annoyed.</i>
7	Working	On-site support needed. Potential failure effect: inoperable subsystems or degradation of main systems, and main product functionality is lower than Customer expectations (for example: all CTQ coming from persona profile which severity level was primary ranked @ 1 or 4) <i>Customer dissatisfaction is experienced</i>	On-site support needed. Potential failure effect: Primary (must have) delivery offer functions degradation (Examples: Missing Quantity/Parcel at Customer, Missing documentation/label, Wrong reference delivered, Product packaging or product damaged but not affecting safety). <i>Customer dissatisfaction is experienced.</i>
9	Compliance/ Property damage	Major Customer business impact: destruction of part of installation or startup impossible. Potential failure effect: non-conformance to regulation (risk of prosecution or huge cost) or inoperable product at the highest system levels, but is not safety related. <i>Customer is highly dissatisfied</i>	Major Customer business impact or Compliance issue. Potential failure effect: Non-conformance to regulation depending country (Examples: Country of origin issue, Regulation related country specific label issue, Invoice issue). <i>Customer is highly dissatisfied.</i>
10	Safety - Human injury	Potential safety issue only. Potential failure effect: injury or harm to the health of human beings. This includes items of non-conformance to regulation representing direct potential safety problem. <i>Customer is endangered.</i>	Potential safety issue only. Potential failure effect: Injury or harm to the health of human beings. This includes items of non-conformance to regulation representing direct potential safety problem. (Examples: Damaged product delivered with safety risk - fall from >70 cm in packaging or any product without packaging, Shipment falls on customer from truck - Cargo Securing issue). <i>Customer is endangered.</i>

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Process Failure Modes Effects Analysis (PFMEA) Procedure

Appendix B: Occurrence Table

Occurrence			
OCC	Level	Prevention Control	
		High Volume (Large quantities of standardized products with repeatable/automated flow to achieve economies of scale (continuous flow))	General Example
		Low Volume (Small batches, higher mix, low manufacturing frequency, high cycle time; often ETO, ECOFIT, adaptation center, spare parts)	
1	Almost uncertain	System to prevent the occurrence, with interlocking	1. POKA-YOKE (Design or Process) 2. Operation with interlocked operator assistance on one by one part (for example pick to light systems / MES /Scanning ... WITH interlocking)
2	Very Low	System/Operation with technology assistance to prevent the occurrence, without interlocking AND Probability of defect is almost uncertain $Cpk \geq 1.66$ or $PPM < 1PPM$	System/Operation with technology assistance to prevent the occurrence, without interlocking OR Manual operations without interlocking (5 prevention actions with 2 actions for maintainability) AND (applies to all above) Probability of defect is almost uncertain (< 1 defect per year)
			High Volume 1. Operation with operator assistance (for example virtual operator assistance/proximity sensors, auto feeder) ... but WITHOUT interlocking) and $Cpk \geq 1.66$ Low Volume (for application refer to GL-GUI-028 PFMEA Scoring rules applied to failure modes) 1) detailed OWS with clear steps, guide & OK/NOK limits (reference, position, tool, ...) 2) qualified for technologies for autonomous operation / Level 2 and above (Technology qualification 1.2.3) 3) certified for autonomous station-based operations / Level L and above (Versatility certification L,U,O) 4) picking assistance (station design, pick to light, ...) 5) operation assistance (validated steps) with maintainability with : 6) self inspection 7) LPA specific question on failure mode and probationary period of extra detection method of maximum of 6 months (TQW mindset : during process or TQW)
4	Low	Operation with technology assistance to reduce human error and improve consistency OR Probability of defect is low $1.33 < Cpk \leq 1.66$ or $PPM < 64 ppm$	Operation with technology assistance to reduce human error and improve consistency OR Probability of defect is low (< 1 defect per month)
7	Medium	Human dependent with control of the process OR Probability of defect is medium $1 < Cpk \leq 1.33$ or $PPM < 2700 ppm$	Human dependent with control of the process OR Probability of defect is medium (< 1 defect per day)
10	Almost certain	Human dependent with no control of the process OR The capability is not known or $Cpk \leq 1$	Human dependent with no control of the process Nothing

General

Process Failure Modes Effects Analysis (PFMEA) Procedure

Appendix C: Detection Table

Detection			
DET	Level	Likelihood to Detect Cause and/or Failure Mode	
		High Volume (Large quantities of standardized products with repeatable/automated flow to achieve economies of scale (continuous flow))	Low Volume (Small batches, higher mix, low manufacturing frequency, high cycle time; often ETO, ECOFIT, adaptation center, spare parts)
1	Almost certain Will not allow non-conformances to escape.	Detection methods proven to always detect the failure mode or failure cause, with interlocking. AND GRR <= 10% or Attribute GRR > =90%	
2	Very High Probably will not allow non-conformances to escape.	Detection methods proven to always detect the failure mode or failure cause without interlocking. AND GRR <= 10% or Attribute GRR > =90%	Detection methods proven to always detect the failure mode or failure cause without interlocking. OR MANUAL: 200% attribute checks with marking and records for failure modes needing operator qualification level >=2 in technology OR MANUAL: 300% attribute checks with records for failure modes without technology expertise (like missing component) AND (GRR applies to all above)
4	Medium Unlikely to allow non-conform	Manual: 100% Attribute checks with records for failure modes, or 200% attribute checks without records, or sampling with variable data SPC OR Automated or Semi-automated detection with alerts or inspection equipment for failure mode AND (GRR applies to all above) GRR<=20% or Attribute GRR>=80%	
7	Low Likely to allow non-conforming product to escape	Manual: 100% Attribute checks without records, or sampling without variable data SPC OR Semi-automated detection with alerts or inspection equipment for failure mode (including sampling). OR GRR<=30%, Attribute GRR >= 70%	
10	Almost uncertain	Failure mode and cause are not detected or cannot be detected OR GRR > 30%, Attribute GRR < 70%	

General

REVISION HISTORY SUMMARY

Approvers	Requested By	Change Description	Reason for Change	Revision	Revision Date	Contributors
ERIC JACQUOT	GURDIP PANESAR	Updated Figure 1 to reflect the revised PFMEA process flow, now incorporating Reverse FMEA. Clarified guidance on PFMEA team structure, including core and extended team roles. Removed the Reliasoft tool requirement and clarified the use of the ForQ_GL33 template. Refined the definition of a failure mode to align with failure to meet product characteristics or functional requirements. Added clarification on causes of failure modes, specifically for mis-assembly and mis-zoning scenarios. Updated the End Effects section to emphasize that PFMEAs focus solely on external end effects, consistent with DFMEA. Internal end effects are addressed through safety audits during Pilot Run/PPAP IG. Clarified RPN matrix rules and governance within the Monitoring and Control section of the procedure. Updated Appendix with latest S, O, D	New PFMEA Occurrence & Detection grids	2.0	11/14/2025	BEN GANESHAN;OMAR GHANI;BERTRAND COURQUIN;SUSANNA CABOS;HAOYING ZHOU;MUGE DEMIRBAS;JEROME FIEL;DAVID MARTIN;SERGE LARDINOIS;ZOLTAN SZABO;YIGIT BAKIS;PATRICK JEAN MICHEL GOUJON;ARTHUR BELHACHE

		grids				
MAGDALENA MICHULEC; ERIC JACQUOT	GURDIP PANESAR	* This document has been migrated from the Global QMS Documents Platform (SharePoint) without revision changes. Process owner/Document author and revision details are specified in the document itself. * A summary of previous revisions is available only in the migrated version.	Document migration	1.1	02/29/2024	